AIT

The Intent Engine Behind AID

AI Can Talk. But Can It Operate?

- Most AI systems generate text, not action
- Earth Observation workflows are complex, fragmented, expert-heavy
- Users ask practical questions, not technical pipeline commands
- AID needs a reliable layer between natural language and execution
- That missing layer is **AIT**

AIT — Executable Intent Object

Turning user questions into executable Earth Observation workflows

- A new intent-aware container for AID
- Stores executable intent-to-task queries
- Connects user questions to the right data, models, and workflows
- Turns natural language into structured, executable tasks
- Learns from execution and improves routing over time
- Enables faster, more consistent, and scalable decision pipelines

AIT — Executable Intent Object

- Stores resolved meaning of the user request
- Links intent to candidate tasks and workflows
- Ranks data sources and models with weights
- Preserves execution state and provenance
- Learns from every successful or failed run

User Query → AIT Object → Workflow Engine → Output

JSON


```json
{
 "query":"Which parcels are unsafe?",
 "intent":"land_use_planning",
 "task":"parcel_multi_risk_screening",
 "status":"resolved"
}
```

Inside One AIT Object

```
AIT
├── Header
├── Query Layer
├── Intent Layer
├── Embedding Layer
├── Task Layer
├── Data Links
├── Workflow Candidates
├── Weights
├── Outputs
└── Update History
```

- Modular and hierarchical
- Extendable over time
- Supports reproducibility
- Designed for explainability

AIT Thinks in Networks, Not Lists

```
subsidence_risk
  ├─ InSAR workflow (0.92)
  ├─ DEM slope layer (0.71)
  ├─ Groundwater signal (0.63)
  └─ Asset exposure map (0.88)
```

- Intent linked to tasks, data, models
- Weighted relationships guide routing
- Weights updated after outcomes
- Supports adaptive intelligence over time

AIT v1 — Ready to Build

Best Practical Architecture Now

```
Frontend (AID UI)
↓
Intent Resolver (LLM + Rules)
↓
AIT Object Store (JSON / SQLite)
↓
Workflow Registry
↓
Routing Engine
↓
EO Execution Layer
↓
Results / Dashboards
```

- Fast to prototype
- Low engineering risk
- Immediate value to AID
- Can scale later to .ait

AIT in Action Across Sectors

Example Use Cases

- Smart Cities


```
{
 "query":"Where should we inspect first?",
 "task":"inspection_priority_screening"
}
```


- Forestry


```
{
 "query":"Where is forest stress increasing?",
 "task":"forest_change_alert"
}
```


- Risk/Insurance


```
{
 "query":"Which parcels combine flood and subsidence risk?",
 "task":"parcel_risk_scoring"
}
```

Roadmap & Funding Case

From Internal Engine to Strategic Platform

Phase 1
AIT inside AID
(JSON objects + routing)

Phase 2
Adaptive graph +
embeddings +
feedback learning

Phase 3
Portable `.ait` containers + APIs + SDK

Phase 4
Cross-industry standard for intent-native workflows

Funding Ask

- Product engineering
- Workflow integration
- Learning engine
- Pilot deployments